

SmartGreenAI: Cilantro Crop

IoT-Based Precision Agriculture and AI Decision-Support System for Greenhouse Cilantro Production

Contenido

1. Project Overview	3
2. General tasks	4
3. Project Scope.....	5
3.1 IoT layer [sensors]	5
3.2 Cloud layer.....	5
3.3 Application layer	5
3.4 Artificial Intelligence layer.....	5
5. System components. 5 Layers.....	5
5.1 Architecture.....	6
Layer 1 Physical/IoT.....	6
Layer 2 Actuation Layer	6
5.2. Communication	6
5.3. Cloud Data Storage.....	6
6. User Roles.....	6
6.1 Agricultural User / Cilantro producer.....	6
Functionalities	6
8.2 Administrator	7
8.3 Super Administrator	8
9. Functional Requirements	8
Authentication and Authorization	8
IoT Monitoring.....	8
Alerts	8
Actuators	9
Cloud Data	9
Artificial Intelligence.....	9
10. Non-Functional Requirements	9
11. Artificial Intelligence Component.....	10

AI functions	10
12. Main Use Cases	10
15. Sprint 0 Project Initiation and Architecture	11
Main Activities	11
Main Deliverables.....	11
16. Sprint 1 IoT and Cloud Monitoring.....	12
User Stories	12
Main Deliverables.....	12
17. Sprint 2 Dashboard, Alerts and Actuators.....	12
User Stories	12
Main Deliverables Sprint 2	13
18. Sprint 3 Artificial Intelligence and Decision Support.....	13
User Stories	13
Main Deliverables.....	14
19. Sprint 4 Integration, Validation and Final Deployment.....	14
User Stories	14
Main Deliverables.....	14
22. Proposed System Workflow	15
24. Acceptance Criteria for the Final Product	15

1. Project Overview

SmartGreenAI: Cilantro Crop will be a precision agriculture system designed to monitor and optimize the environmental and irrigation conditions of a greenhouse for Cilantro production. The system will combine an IoT sensor network, actuators, cloud computing, data visualization, alerts, and Artificial Intelligence (AI) to help agricultural producers make better decisions.

The system will continuously collect information from the greenhouse, including:

- Temperature
- Humidity
- Light intensity
- CO₂ concentration
- Water-tank level
- pH, if available

The IoT devices will transmit the information to a cloud server, where the data will be stored and processed. Based on predefined thresholds and AI-based analysis, the system will generate alerts and recommendations for the Cilantro producers.

The system will also control selected agricultural actuators, such as:

- Irrigation pump
- Greenhouse ventilation fans
- Shade system
- Auxiliary lighting, if available

The objective is not to completely replace the Cilantro producer's decisions, but to provide real-time information and intelligent recommendations that support irrigation, environmental control, and crop management.

SmartGreenAI: Cilantro Crop will be a cloud-connected precision-agriculture platform for Cilantro greenhouse management. A network of IoT sensors will continuously collect environmental and agricultural information. The information will be transmitted to a cloud platform, stored in a historical database, analyzed, and presented through a web dashboard and displays.

The platform will combine traditional threshold-based alerts with Artificial Intelligence. While conventional rules can detect immediate problems, the AI component will analyze multiple variables and historical patterns to generate more contextual recommendations.

The three user roles Agricultural User, Administrator, and Super Administrator —will provide an appropriate separation between agricultural operation, technical administration, and platform-level administration.

The final prototype will demonstrate the complete cycle:



With 5 team members working 5 hours per week for 14 weeks, the project provides a 350-hour academic software engineering project that can be managed using Scrum through five incremental sprints.

2. General tasks

1. Design an IoT network capable of collecting greenhouse environmental and Cilantro agriculture data.
2. Implement communication between IoT devices and a cloud server.
3. Store historical sensor information in a cloud database.
4. Develop a web-based monitoring dashboard.
5. Implement threshold-based alerts.
6. Implement remote or automatic control of selected actuators.
7. Develop an AI component capable of analyzing historical and real-time information.
8. Generate AI-based recommendations for Cilantro producers.
9. Provide different functionalities according to the user's role.
10. Validate the complete system using simulated and/or real sensor data.

3. Project Scope

The prototype will include:

3.1 IoT layer [sensors]

- Microcontroller/IoT gateway
- Communication protocol
- Actuators
- Local data acquisition

3.2 Cloud layer

- API/backend
- Cloud database
- Authentication
- Data processing
- Historical data storage

3.3 Application layer

- Web dashboard
- Sensor visualization
- Historical data
- Alerts
- Actuator control
- User management

3.4 Artificial Intelligence layer

- Data analysis
- Detection of abnormal conditions
- Identification of trends
- Irrigation/environmental recommendations
- Natural-language explanation of recommendations

5. System components. 5 Layers

5.1 Architecture

Layer 1 Physical/IoT

- Sensors collect information from the greenhouse.
 - Temperature
 - Humidity
 - Light
 - Sensors may include:
 - CO₂
 - Water level
 - pH
 - Other

Layer 2 Actuation Layer

- The system will control:
 - Irrigation pump
 - Ventilation fans
 - Optional greenhouse lighting

5.2. Communication

The IoT devices should communicate with the cloud through an Internet connection.

5.3. Cloud Data Storage

All relevant measurements must be stored in a cloud server.

- Sensor ID
- Greenhouse ID
- Measurement type
- Measurement value
- Unit
- Timestamp
- Device status

6. User Roles

The system will have three user roles.

6.1 Agricultural User / Cilantro producer

The agricultural user is the primary operational user.

Functionalities

- Login
- View greenhouse status

- View current sensor measurements
- View historical measurements
- View graphs
- View alerts
- Receive irrigation alerts
- Receive environmental alerts
- View AI recommendations
- Accept/reject recommendations
- Manually activate/deactivate irrigation
- Automatic activate/ deactivate irrigation
- View actuator status
- Select automatic/manual operating mode
- View historical irrigation events
- Register observations about the crop

Example:

"Soil in Area 2 is below the recommended range. Irrigation is recommended for approximately 8 minutes."

8.2 Administrator

The administrator manages the operational configuration of the platform.

Functionalities

- Login
- Manage agricultural users
- Register greenhouses
- Register cultivation areas
- Register kind of crop
 - Automatic configuration parameters like: humidity, light, etc depends of kind of crop
- Register IoT devices
- Register sensors
- Register actuators
- Configure sensor thresholds
- Configure alert rules
- Configure irrigation parameters
- Monitor IoT device connectivity
- View system logs
- View system statistics
- Configure notification mechanisms

8.3 Super Administrator

The super administrator manages the entire platform.

Functionalities

- Create administrators
- Modify administrators
- Deactivate users
- Manage roles and permissions
- Configure global system parameters
- Manage greenhouse organizations/tenants
- View global system logs
- Configure security parameters
- Manage AI configuration
- Manage system integrations
- Monitor overall system availability

9. Functional Requirements

Authentication and Authorization

- FR-01. The system shall allow users to authenticate using username/email and password.
- FR-02. The system shall assign permissions according to the user's role.
- FR-03. The system shall prevent unauthorized users from accessing restricted functionality.
- FR-04. The system shall allow administrators to activate/deactivate users.

IoT Monitoring

- FR-05. The system shall receive measurements from IoT devices.
- FR-06. The system shall identify each sensor uniquely.
- FR-07. The system shall associate each sensor with a greenhouse area.
- FR-08. The system shall store sensor measurements in the cloud.
- FR-09. The system shall display current sensor measurements.
- FR-10. The system shall display historical sensor measurements.
- FR-11. The system shall indicate when an IoT device is disconnected.

Alerts

- FR-12. The system shall compare sensor measurements against configured thresholds.
- FR-13. The system shall generate alerts when measurements exceed predefined thresholds.
- FR-14. The system shall classify alerts according to severity (Slow,Medium,High).
- FR-15. The system shall maintain an alert history.

Actuators

- FR-16. The system shall monitor actuator status.
- FR-17. Authorized users shall be able to activate/deactivate actuators manually.
- FR-18. The system shall support automatic actuator activation according to predefined rules.
- FR-19. The system shall record actuator activation events.

Cloud Data

- FR-20. The system shall store historical sensor data.
- FR-21. The system shall store user activity.
- FR-22. The system shall store alerts.
- FR-23. The system shall store actuator events.
- FR-24. The system shall provide APIs for IoT devices and the web application.

Artificial Intelligence

- FR-25. The system shall analyze historical sensor data.
- FR-26. The system shall identify trends in environmental conditions.
- FR-27. The system shall identify abnormal measurements.
- FR-28. The system shall generate recommendations for the agricultural user.
- FR-29. The system shall provide an explanation for each AI recommendation.
- FR-30. The system shall consider multiple variables when generating recommendations.

10. Non-Functional Requirements

- NFR-01 Performance. Normal dashboard operations should respond within approximately 3 seconds under the expected prototype workload.
- NFR-02 Scalability. The architecture should allow additional sensors, greenhouse areas, and greenhouses to be added without redesigning the complete system.
- NFR-03 Security. Passwords shall not be stored in plain text.
- NFR-04 Authorization. Users shall only access functionality authorized for their role.
- NFR-05 Data Integrity. The system shall maintain timestamps and sensor identifiers for all measurements.
- NFR-06 Usability. The dashboard should provide clear graphs, status indicators, and alerts that can be understood by a non-technical agricultural user.
- NFR-07 Maintainability. The system should be implemented using modular components so that IoT, backend, frontend, database, and AI components can evolve independently.
- NFR-08 Reliability. The system shall detect missing or abnormal sensor data.

- NFR-09 Security of IoT Communication. IoT-to-cloud communication should use authentication and encrypted communication when supported by the selected infrastructure.

11. Artificial Intelligence Component

- The AI component is one of the principal differentiators of SmartGreenAI: Cilantro Crop.
- The initial AI implementation can focus on decision support rather than complete autonomous control.

AI functions

- Environmental trend analysis
- Anomaly detection
- Irrigation recommendation
- Risk identification
- Production-condition analysis
- Natural-language recommendations

Example. The system detects: “Temperature: 31 °C, Humidity: 48 %, Soil moisture: 27 %, Last irrigation: 18 hours ago, Forecast condition: High temperature

The AI may generate: Irrigation should be considered for Area 2 because soil moisture is below the configured target and the current temperature is high. Verify water availability before activating the irrigation system.

The AI recommendation should include: Recommendation, Reason, Relevant sensor measurements, Confidence or certainty indicator, when available Timestamp,

12. Main Use Cases

- UC01 Authenticate All users
- UC02 Monitor greenhouse Agricultural user
- UC03 View sensor history Agricultural user
- UC04 View alerts Agricultural user
- UC05 View AI recommendations Agricultural user
- UC06 Control irrigation Agricultural user
- UC07 Manage sensors Administrator
- UC08 Configure thresholds Administrator
- UC09 Manage greenhouse Administrator
- UC10 Manage users Administrator
- UC11 Manage administrators Super administrator
- UC12 Configure global parameters Super administrator
- UC13 Monitor system logs Administrator/Super administrator

- UC14 Analyze agricultural data System/AI
- UC15 Generate recommendation AI/System

15. Sprint 0 Project Initiation and Architecture

Objective Define the project architecture, requirements, technology stack, backlog, IoT prototype, data model, and development environment.

Main Activities

- Define product vision
- Define personas
- Define user roles
- Define functional requirements
- Define non-functional requirements
- Define system architecture
- Select sensors
- Select microcontroller
- Define communication protocol
- Design database
- Design initial UI
- Configure Git repository
- Configure development environment
- Create initial IoT proof of concept
- Define Scrum backlog

User Story Syntax → As a <role>, I want to <need> so that <objective>

Main Deliverables

- Project charter
- Product vision
- Product backlog
- System architecture
- Requirements specification
- Database model
- UI prototype
- Initial IoT prototype
- Git repository
- Development environment
- Definition of Done

16. Sprint 1 IoT and Cloud Monitoring

Objective: Implement the basic IoT network and cloud infrastructure capable of receiving and storing greenhouse: Cilantro measurements.

User Stories

- US101 Sensor acquisition. As a system, I want to collect temperature and humidity measurements so that greenhouse environmental conditions can be monitored.
- US102 Soil moisture. As a Cilantro producer, I want soil moisture measurements to be collected so that I can determine whether irrigation may be required.
- US103 Cloud transmission . As a system, I want IoT measurements to be transmitted to the cloud so that information can be stored remotely.
- US104 Historical storage . As a Cilantro producer, I want measurements to be stored with timestamps so that I can analyze greenhouse conditions over time.
- US105 Device status. As an administrator, I want to know whether an IoT device is connected so that I can identify communication problems.

Main Deliverables

- Functional IoT prototype
- Temperature/humidity monitoring
- Soil moisture monitoring
- Cloud communication
- Cloud database
- IoT infrastructure
- Sensor registration mechanism
- Device-status monitoring
- Initial technical documentation

Sprint Review Demonstration

Sensor→ESP32 → Internet →Cloud→ Database. (with real-time measurements visible in the cloud)

17. Sprint 2 Dashboard, Alerts and Actuators

Objective: Develop the web application for monitoring the greenhouse:Cilantro and implement alert and actuator functionality.

User Stories

- US201 User authentication. As a user, I want to log into the system so that I can access the functions authorized for my role.
- US202 Greenhouse:Cilantro, dashboard. As a Cilantro producer, I want to see current sensor measurements on a dashboard so that I can understand the current greenhouse conditions.
- US203 Historical graphs. As a Cilantro producer, I want to visualize historical sensor data so that I can identify environmental trends.

- US204 Threshold alerts. As a Cilantro producer, I want to receive an alert when environmental conditions exceed acceptable limits so that I can take corrective action.
- US205 Irrigation control. As a Cilantro producer, I want to manually activate irrigation so that I can respond to a specific crop condition.
- US206 Automatic irrigation. As a Cilantro producer, I want the system to activate irrigation automatically according to predefined conditions so that irrigation can be managed more efficiently.
- US207 Sensor configuration. As an administrator, I want to configure sensors and thresholds so that the system can be adapted to the greenhouse.

Main Deliverables Sprint 2

- Login
- Role-based access
- Cilantro producer dashboard
- Administrator dashboard
- Real-time sensor visualization
- Historical graphs
- Alert engine
- Alert history
- Irrigation actuator
- Manual actuator control
- Automatic actuator rules
- Sensor configuration
- Actuator event history

18. Sprint 3 Artificial Intelligence and Decision Support

Objective: Implement the AI component capable of analyzing agricultural information and generating recommendations.

User Stories

- US301 Historical analysis. As a Cilantro producer, I want the system to analyze historical sensor information so that I can understand environmental patterns.
- US302 Anomaly detection. As a Cilantro producer, I want the system to identify unusual environmental conditions so that potential problems can be detected early.
- US303 Irrigation recommendation. As a Cilantro producer, I want AI to recommend when irrigation may be necessary so that I can make better irrigation decisions.
- US304 Multi-variable analysis. As a Cilantro producer, I want AI to consider temperature, humidity, soil moisture, and historical irrigation simultaneously so that recommendations are based on multiple factors.

- US305 Explainable recommendation. As a Cilantro producer, I want each AI recommendation to include an explanation so that I understand why the recommendation was generated.
- US306 AI configuration. As an administrator, I want to configure the parameters used by the AI component so that the system can be adapted to the greenhouse.

Main Deliverables

- AI data-processing pipeline
- Data preprocessing
- Exploratory agricultural→data analysis
- Irrigation recommendation engine
- AI recommendation dashboard
- Recommendation explanation
- AI technical documentation

19. Sprint 4 Integration, Validation and Final Deployment

Objective: Integrate all components and validate the complete precision-agriculture Cilantro platform.

User Stories

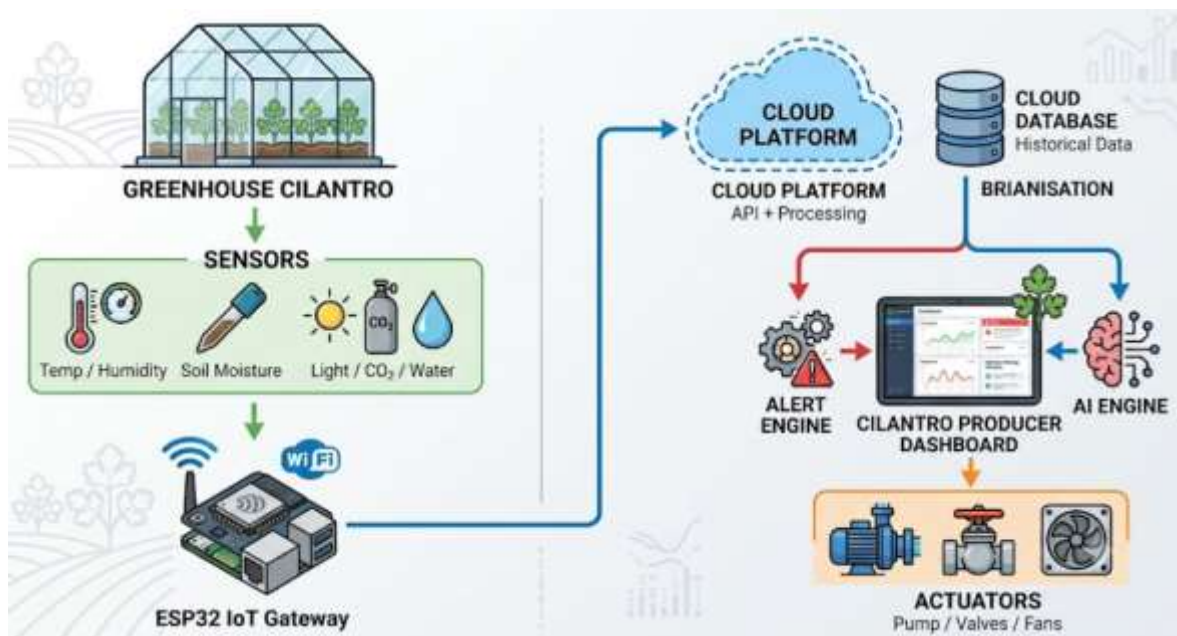
- US401 Integrated monitoring. As a Cilantro producer, I want sensor information, alerts, actuator status, and AI recommendations in one dashboard so that I can manage the greenhouse from a single platform.
- US402 End-to-end operation. As a Cilantro producer, I want sensor measurements to automatically reach the cloud and generate alerts/recommendations so that the complete system operates continuously.
- US403 System administration. As a super administrator, I want to manage administrators and global configuration so that the platform can be maintained securely.
- US404 Reliability . As an administrator, I want the system to identify disconnected sensors so that communication failures can be addressed.
- US405 Validation . As a project team, we want to execute system tests so that the system's functional and non-functional requirements can be validated.
- US406 Final demonstration . As a stakeholder, I want to see an end-to-end demonstration so that I can evaluate the value of the SmartGreenAI: Cilantro Crop solution.

Main Deliverables

- Integrated SmartGreenAI: Cilantro Crop platform

- Complete IoT network
- Cloud infrastructure
- Database
- Web application
- Alert system
- Actuator control
- AI recommendation system
- Security implementation
- System testing
- User manual
- Technical documentation
- Final presentation
- Final demonstration

22. Proposed System Workflow



24. Acceptance Criteria for the Final Product

The project will be considered successful if the team can demonstrate the following scenario:

- A soil-moisture sensor measures a low moisture value.
- The IoT device transmits the measurement to the cloud.
- The cloud stores the measurement.
- The dashboard displays the measurement.
- The alert engine identifies the low-moisture condition.

- An alert is generated.
- 7. The AI analyzes the current and historical data.
- 8. The AI generates an irrigation recommendation.
- 9. The Cilantro producer receives the recommendation.
- The Cilantro producer can activate irrigation manually.
- Alternatively, automatic irrigation can be activated according to configured rules.
- The actuator event is recorded.
- New sensor measurements are collected.
- The system verifies whether soil moisture has improved.
- The complete process is visible in the dashboard.